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SafeDojo: Safe Reinforcement Learning for VLA via Interactive World Model

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Abstract: Safe control is a prerequisite for real-world embodied intelligence, for which safe reinforcement learning has emerged as a promising paradigm. However, existing safe reinforcement learning methods either require costly real-world exploration or depend on hand-crafted safety functions. Neither scales to vision-language-action models deployed in open-world physical environments. We propose **SafeDojo**, a model-based safe reinforcement learning framework that improves vision-language-action policies through world-model imagination. SafeDojo performs online reinforcement learning on top of an interactive video world model. A ResNet success classifier scores predicted future frames, while a lightweight safety head predicts per-step contact probabilities from the current latent context and proposed action chunk, jointly assessing task execution and trajectory safety. These decoupled signals are balanced through Lagrangian-constrained GRPO. On SafeLIBERO Level I, SafeDojo improves average safe-success rate by 8.25 percentage points over the strongest baseline; on the zero-shot Level II setting, it is competitive with the strongest model-free baseline. Real-world Franka deployment further shows the best average task and safe-success rates across five tasks. These results support world-model-based safe reinforcement learning as a promising path toward safer embodied intelligence.

Keywords: Safe Reinforcement Learning, Vision-Language-Action, World Model

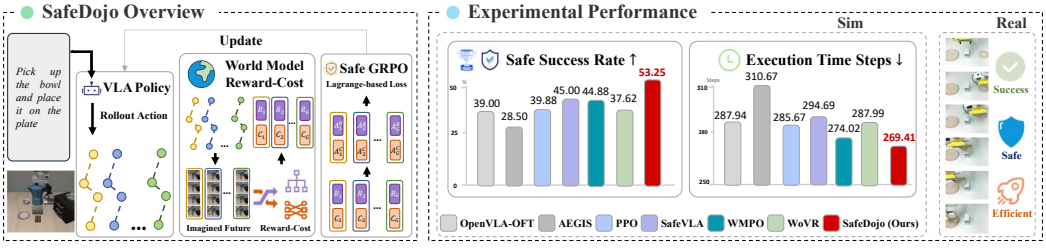


Figure 1: **Overview of SafeDojo.** SafeDojo enhances VLA policies with world-model-based reward–cost evaluation and safe GRPO, boosting safe success and efficiency in simulation and real-world scenarios. The performance panels compare against all baselines; complete comparisons are reported in Tables 1 and 2.

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1 Introduction

Safe control is an urgent challenge for deploying autonomous robots in real-world environments [1, 2]. Among various safety concerns, collision avoidance during contact-rich robotic manipulation stands out as one of the most critical, since even a single violation can cause task failure, irreversible hardware damage or environmental disruption [3, 4, 5, 6, 7, 8, 9]. As vision-language-action (VLA) models become the dominant paradigm for general-purpose robotic manipulation [10, 11, 12, 13, 14], we focus on a key open problem, ***learning collision-free safe VLA policies in obstacle-rich environments***. Unlike low-dimensional control with explicit or compactly learned safety constraints [15, 16], VLA policies must infer safety boundaries from high-dimensional visual observations across diverse and previously unseen obstacle configurations [17]. This challenge is compounded by the irreversibility of physical violations, demanding that unsafe actions be anticipated and prevented before execution. Existing approaches to safe robotic manipulation span several paradigms, including inference-time control barrier functions [18], model-free constrained reinforcement learning (RL) [17], and model-based safe RL [16, 19]. Control barrier functions provide theoretical safety guarantees but require precise geometric modeling and camera calibration that are impractical for open-world VLA deployment. Model-free methods enforce safety through penalty-based optimization but cannot anticipate violations before they occur. Model-based safe RL addresses this limitation by performing anticipatory rollouts in imagination. However, existing methods largely operate in low-dimensional state spaces with simulator-provided cost functions, neither of which directly transfers to VLA models. Extending model-based safe reinforcement learning to VLA is particularly challenging because task progress and collision avoidance are inherently conflicting objectives, and safety signals must be learned from high-dimensional visual observations rather than analytically defined. To our knowledge, model-based safe reinforcement learning for vision-language-action models remains unexplored. We propose SafeDojo as the first framework to address it.

We propose SafeDojo, to our knowledge the first world model-based safe RL framework for VLA models. SafeDojo is motivated by **a central challenge in safe VLA learning, where policy improvement requires exploration, yet a single unsafe contact in the real world can cause task failure, hardware damage, or environmental disruption.** SafeDojo performs online policy optimization entirely within an interactive video world model, avoiding risky real-world exploration. The world model rolls out candidate trajectories into action-conditioned future video frames and latent representations, from which a dual-branch evaluator estimates dense task reward and safety cost signals. Specifically, a finetuned ResNet success classifier predicts per-step task progress from imagined frames, while a lightweight but effective safety head estimates per-step safety-cost signals from the current latent context and the proposed action chunk. These dense signals are aggregated into trajectory-level reward-cost pairs and optimized with a Lagrangian-based safety-constrained GRPO objective. This design enables SafeDojo to improve task success and collision avoidance under explicit safety constraints, without hand-crafted safety functions or risky real-world trials.

Our contributions are summarized as follows. **(1) World Model based Reward-Cost.** We develop an interactive video world model for safe VLA rollouts, together with a dual-branch evaluator that estimates decoupled task reward and safety cost from imagined dynamics. **(2) Safety-Constrained RL.** We introduce Lagrangian-based safety-constrained GRPO to optimize policies under explicit safety constraints, avoiding fixed penalty while coordinating task success and collision avoidance. **(3) Superior Empirical Performance.** SafeDojo achieves strong gains on SafeLIBERO and real-world manipulation, improving the Level I average safe-success rate by 8.25 percentage points over the strongest baseline while achieving the best average task success and execution efficiency on SafeLIBERO.

2 Related Work

Existing VLA safety mechanisms rely on perception augmentation [20, 21], inference-time intervention [18, 22, 23], or policy-level alignment [17, 24], but do not optimize VLA policies with model-based safe RL. Safe RL under Constrained Markov Decision Processes (CMDPs) [25] offers a principled formulation, yet model-free methods [26, 27, 28, 29, 30] require costly and unsafe

exploration, while model-based methods [31, 16, 32, 19, 33, 34] mostly target generic control with predefined scalar costs. World-model-based VLA post-training [35, 36, 37, 38, 39] optimizes task return in imagination but lacks safety constraints. Detailed related work is provided in Appendix B.

3 Problem Definition and Preliminaries

3.1 Problem Formulation

Extending VLA MDPs to CMDPs. Standard VLA can be modeled as a Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$, where $s_t \in \mathcal{S}$ denotes vision-language states, $a_t \in \mathcal{A}$ is the action space, and $r : \mathcal{S} \times \mathcal{A} \to \{0, 1\}$ is the task reward indicator. P, ρ_0, and γ follow standard MDP notation. We extend this formulation to a Constrained MDP (CMDP) [25] $\mathcal{M}^c = (\mathcal{S}, \mathcal{A}, P, r, c, \rho_0, \gamma)$ by introducing a safety cost $c : \mathcal{S} \times \mathcal{A} \to [0, 1]$, requiring the VLA policy π_θ to maximize $J^r(\pi_\theta) = \mathbb{E}_{\pi_\theta}[\sum_t \gamma^t r_t]$ subject to $J^c(\pi_\theta) = \mathbb{E}_{\pi_\theta}[\sum_t \gamma^t c_t] \leq d$, where d denotes the allowable safety budget.

Model-based Safe RL for VLA. Under this CMDP formulation, model-based safe reinforcement learning for VLA improves the policy using imagined trajectory feedback, rather than relying on static safety rules, post-hoc action filtering, or risky real-world exploration. Each policy rollout induces a trajectory τ with task rewards and safety costs accumulated over time. These accumulated signals are converted into advantages that drive policy-gradient updates of the form $\nabla_\theta \log \pi_\theta(a_t|s_t)A_t$, where task rewards encourage goal-directed actions and safety costs suppress collision-prone ones. τ is considered safe-successful only if it completes the instructed task without incurring collision cost, making safe-success rate a stricter criterion than task success alone.

3.2 Preliminaries

Our method builds on Group Relative Policy Optimization (GRPO), which optimizes a policy using relative feedback among multiple sampled trajectories without learning a separate value critic. For a given state, the old policy π_θ^{old} samples a group of G trajectories $\{\tau_i\}_{i=1}^G$ and obtains scalar rewards $\{R_i\}_{i=1}^G$. GRPO normalizes rewards within the group to obtain $\hat{A}_i = (R_i - \mu_G(R))/(\sigma_G(R) + \epsilon)$ and updates the policy with the clipped objective:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_{i,t}\left[\min\left(\rho_{i,t}(\theta)\hat{A}_i, \text{clip}\left(\rho_{i,t}(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_i\right)\right], \tag{1}$$

where $\rho_{i,t}(\theta) = \pi_\theta(a_{i,t}|s_{i,t})/\pi_\theta^{\text{old}}(a_{i,t}|s_{i,t})$ is the probability ratio. This formulation reinforces above-average samples and suppresses below-average ones, making it suitable for VLA optimization.

4 Method

Building on the safe VLA policy optimization problem defined in Sec. 3.1, we propose **SafeDojo**, to our knowledge the first model-based safe reinforcement learning framework for VLA policy improvement. SafeDojo is motivated by the core tension between policy exploration for task improvement and unsafe contacts that may cause task failure, hardware damage, or environmental disruption. As illustrated in Fig. 2, SafeDojo addresses this challenge through three key innovations, including world model based virtual safety exploration in Sec. 4.1, decoupled reward-cost estimation in Sec. 4.2, and Lagrangian-based safety-constrained GRPO in Sec. 4.3. Together, these designs enable VLA policies to improve task success while reducing collision risks, without damage-prone real-world trials.

4.1 Chunk-wise Interactive World Model Rollouts

Instead of collecting policy rollouts in the physical environment, SafeDojo performs model-in-the-loop rollouts inside an interactive world model p_ϕ. We instantiate the interactive world model as an action-conditioned video predictor that generates future visual observations and latent dynamics. Given an initial visual observation o_0 and a language instruction l, the current VLA policy π_θ samples a group of candidate trajectories $\{\tau_i\}_{i=0}^{G-1}$, where each trajectory consists of K action chunks. For an H-step action chunk $a_{i,k,1:H}$, the world model predicts H temporally aligned future frames and

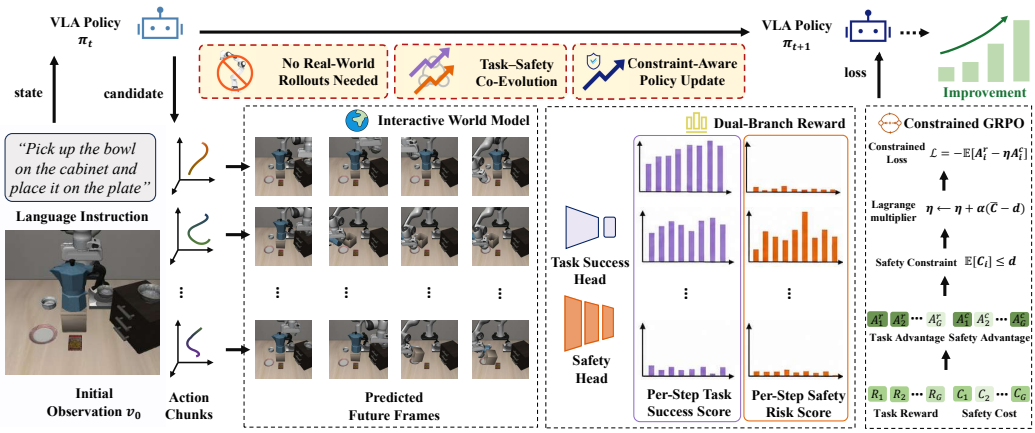


Figure 2: **Detailed SafeDojo Pipeline.** SafeDojo optimizes VLA policies entirely inside an interactive video world model by rolling out candidate action trajectories into imagined future dynamics. Task reward and safety cost are decoupled and optimized via Lagrangian-based constrained GRPO, improving task success while reducing safety risks without potentially damaging real-world rollouts.

latent states:

$$\{(\hat{o}_{i,k,h}, \hat{z}_{i,k,h})\}_{h=1}^{H} = p_\phi(\hat{o}_{i,k,0}, z_{i,k}^{\text{ctx}}, a_{i,k,1:H}, l), \quad k = 0, \ldots, K-1. \tag{2}$$

Here, $z_{i,k}^{\text{ctx}}$ is the latent context available before rolling out chunk k. We build p_ϕ on Wan 2.2 [40] with cross-attention action conditioning. Each action chunk consists of $H = 8$ control steps, where each step is a 7-DoF command (a 6-DoF delta end-effector pose plus a gripper action) following the OpenVLA-OFT action representation and normalized with the suite-specific statistics.

However, chunk-wise prediction introduces a train-test mismatch between clean demonstration observations used during training and self-generated observations used during inference. This mismatch becomes more severe in long-horizon rollouts due to exposure error, where the world model repeatedly conditions on its own imperfect predictions and accumulates errors across chunks. To improve rollout robustness, we adopt a static-video augmentation during world-model training that occasionally freezes the conditioning frames, exposing p_ϕ to degenerate temporal contexts. This augmentation makes the model more tolerant to imperfect intermediate predictions during autoregressive chunk-wise rollout.

4.2 Decoupled Task Reward and Safety Cost Estimation

Given an imagined trajectory $\hat{\tau}_i = \{(\hat{o}_{i,k,h}, \hat{z}_{i,k,h}, a_{i,k,h})\}_{k=0,h=1}^{K-1,H}$, SafeDojo evaluates it with decoupled task-reward and safety-cost signals. This decoupling is essential for safe VLA optimization, since task progress and safety risk are not always aligned. A trajectory may make progress toward the instruction while causing unsafe contact, or remain collision-free but fail to complete the task. Therefore, instead of collapsing task execution and safety into a hand-crafted scalar reward, SafeDojo estimates task reward and safety cost separately, providing structured feedback for subsequent constrained policy optimization.

The success head scores each predicted frame; the safety head uses the current latent context and proposed action chunk to predict H contact probabilities:

$$r_{i,k,h} = f_{\text{task}}(\hat{o}_{i,k,h}, l), \qquad c_{i,k,h} = \left[f_{\text{safe}}(z_{i,k}^{\text{ctx}}, a_{i,k,1:H})\right]_h, \tag{3}$$

Both outputs are aligned at the control-step level, although their inputs differ. We aggregate them into trajectory-level task reward and safety cost:

$$r_i = \frac{1}{KH} \sum_{k=0}^{K-1} \sum_{h=1}^{H} r_{i,k,h}, \qquad c_i = \frac{1}{M} \sum_{c \in \text{TopM}(\{c_{i,k,h}\}_{k,h})} c, \tag{4}$$

where M denotes the number of high-risk steps used for safety aggregation. We use this top-M mean predicted cost as a surrogate for the discounted cumulative CMDP cost, optimizing $\mathbb{E}[c_i] \leq d$ in

Algorithm 1: SafeDojo: Safe RL for VLA via Interactive World Model

Input: VLA policy π_θ, reference policy π_θ^{ref}, world model p_ϕ, task reward head f_{task}, safety cost head f_{safe}, allowable budget d, group size G, multiplier η, multiplier update rate α_η.
Output: Optimized π_θ^* for task success under safety constraints.

Initialize $\pi_\theta^{\text{old}} \leftarrow \pi_\theta$, $\eta \geq 0$, $d = 0.2$, $\alpha_\eta = 0.05$, $G = 16$;
while *not converged* **do**
  Sample (l, o_0);
  $\{\tau_i\}_{i=1}^{G} \sim \pi_\theta^{\text{old}}(\cdot \mid l, o_0)$; ▷ candidate trajectories
  for $i = 1, \ldots, G$ **do**
    $\hat{\tau}_i \leftarrow p_\phi(o_0, l, \tau_i)$; ▷ rollout, Eq. 2
    $\{r_{i,k,h}, c_{i,k,h}\}_{k=0,h=1}^{K-1,H} \leftarrow (f_{\text{task}}, f_{\text{safe}})(\hat{\tau}_i)$; ▷ dense scores, Eq. 3
    $(r_i, c_i) \leftarrow \text{Agg}(\{r_{i,k,h}, c_{i,k,h}\}_{k,h})$; ▷ aggregation, Eq. 4
  $\hat{A}_i^r \leftarrow \text{Norm}_G(r_i)$, $\hat{A}_i^c \leftarrow \text{Norm}_G(c_i)$; ▷ advantages, Eq. 5
  $\eta \uparrow$ if $\frac{1}{G}\sum_{i=1}^{G} c_i > d$, else $\eta \downarrow$; ▷ multiplier, Eq. 7
  $\hat{A}_i^{\text{safe}} \leftarrow \hat{A}_i^r - \eta \hat{A}_i^c$; ▷ safe advantage, Eq. 6
  $\theta \leftarrow \arg\min_\theta \mathcal{L}_{\text{Safe-GRPO}}(\theta)$; ▷ policy update, Eq. 8
  $\pi_\theta^{\text{old}} \leftarrow \pi_\theta$;
return π_θ^*;

imagination. In implementation, we set $M = 16$. The task reward branch f_{task} is instantiated with a ResNet success classifier finetuned on rollout frames with per-frame success labels, while the safety cost branch f_{safe} is a compact convolutional latent encoder followed by an action-conditioned MLP head that outputs a per-step obstacle-contact probability over the proposed action chunk.

The resulting pair (r_i, c_i) provides a reward-cost characterization of each candidate trajectory. A trajectory with high task reward and low safety cost corresponds to successful and safe execution, while a high-cost trajectory is discouraged even if it appears task-progressive. Conversely, a low-cost but low-reward trajectory is safe but ineffective. This decoupled evaluation allows SafeDojo to distinguish these cases and optimize task success and safety through an explicit constrained objective.

4.3 Lagrangian-based Safety Constrained GRPO

A straightforward way to apply GRPO for safe VLA learning is reward shaping, which forms a scalar reward $\tilde{r}_i = r_i - \lambda c_i$ and then applies the vanilla GRPO objective in Eq. 1. This strategy treats safety as a fixed penalty rather than an explicit constraint, making the reward-cost trade-off sensitive to the manually chosen weight λ. More importantly, it does not explicitly enforce a safety boundary, allowing high-reward trajectories to remain favored even when they violate the desired cost budget.

Inspired by Constrained GRPO [41], SafeDojo instead decouples task reward and safety cost, and optimizes them with a Lagrangian-based safety-constrained GRPO objective. Given the reward-cost pair (r_i, c_i) estimated in Sec. 4.2, we compute group-relative reward and cost advantages separately:

$$\hat{A}_i^r = \frac{r_i - \mu_G(r)}{\sigma_G(r) + \epsilon}, \qquad \hat{A}_i^c = \frac{c_i - \mu_G(c)}{\sigma_G(c) + \epsilon}, \tag{5}$$

where $\mu_G(\cdot)$ and $\sigma_G(\cdot)$ denote the mean and standard deviation over the G candidate trajectories sampled for the same instruction-observation context. Unlike reward shaping, which normalizes a pre-combined scalar reward, this separate normalization preserves the relative structure of task reward and safety cost before Lagrangian combination.

To optimize this surrogate-constrained objective, SafeDojo combines the separately normalized reward and cost advantages using a non-negative Lagrange multiplier η:

$$\hat{A}_i^{\text{safe}} = \hat{A}_i^r - \eta \hat{A}_i^c. \tag{6}$$

η is then updated according to the batch-level constraint violation:

$$\eta \leftarrow \left[\eta + \alpha_\eta \left(\frac{1}{G}\sum_{i=1}^{G} c_i - d\right)\right]_+, \tag{7}$$

where d is the budget on the top-M surrogate cost in Eq. 4, α_η is the multiplier learning rate, and $[\cdot]_+$ projects η to be non-negative. When the average safety cost exceeds d, η increases to penalize unsafe trajectories more strongly. Otherwise, it decreases to allow more task-oriented improvement. We set $d = 0.2$ and $\alpha_\eta = 0.05$.

We obtain the final safety-constrained GRPO objective by replacing the vanilla advantage $\hat{A}_i$ in Eq. 1 with $\hat{A}_i^{\text{safe}}$ and adding KL regularization to the reference policy:

$$\mathcal{L}_{\text{Safe-GRPO}} = -\mathbb{E}_{i,t}\left[\min\left(\rho_{i,t}(\theta)\hat{A}_i^{\text{safe}}, \text{clip}\left(\rho_{i,t}(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_i^{\text{safe}}\right)\right] + \beta D_{\text{KL}}\left(\pi_\theta \| \pi_\theta^{\text{ref}}\right), \tag{8}$$

where π_θ^{ref} is the reference policy and β controls the regularization strength. In experiments, we set $\beta = 0$ and rely on the GRPO clipping ratio for the trust-region constraint. Algorithm 1 summarizes training, including world model rollouts, reward-cost estimation, and safety-constrained GRPO.

5 Experiments

5.1 Experiments Setup

Experimental Details. We evaluate SafeDojo on SafeLIBERO [18], a safety-critical benchmark built on LIBERO [42] in Robosuite [43] with a single-arm Franka Emika Panda 7-DoF manipulator. SafeLIBERO includes four suites, Spatial, Goal, Object, and Long, each with four tasks under two obstacle-interference levels. Level I places obstacles near the target object, while Level II places them farther away but along the motion path. We train on Level I only and evaluate on both levels to test generalization, and further validate SafeDojo on 5 real-world Franka tasks.

Baselines. We compare SafeDojo with six baselines across four categories, all built on the same OpenVLA-OFT [44] variant with parallel discrete action-token prediction, and initialized from a shared SFT checkpoint obtained by fine-tuning OpenVLA-OFT on collision-free SafeLIBERO demonstrations. This variant enables token-level log-probability computation for GRPO. OpenVLA-OFT serves as our Safe-SFT baseline: it is supervised-finetuned on the collision-free expert demonstrations, without RL or an additional safety mechanism. AEGIS [18] represents inference-time safety by applying a plug-and-play CBF-based action filter. For model-free RL, we include reward-only PPO and SafeVLA [17], which formulates safety under the CMDP framework; both use the same SFT initialization as SafeDojo. For model-based RL, we compare against WMPO [35] and WoVR [36], recent world-model-based policy optimizers for VLA without explicit safety constraints.

Training Details. For each suite, we use a fixed offline set of $1{,}500$ SafeLIBERO trajectories (750 collision and 750 collision-free) to train world-model and safety components shared by its four tasks. All unsafe supervision comes from simulation; real-robot adaptation uses 50 collision-free demonstrations per task and no unsafe robot data. We finetune the interactive world model from Wan2.2 for $5{,}000$ optimization steps with a learning rate of 1×10^{-5}. The task reward classifier and the safety contact head are each trained for 20 epochs with learning rates of 1×10^{-4} and 3×10^{-4}, respectively. For policy optimization, SafeDojo runs safety-constrained GRPO with group size $G = 16$ and a learning rate of 2×10^{-5}, without collecting additional unsafe trajectories. Appendix H gives compute, checkpoint-granularity, and data-budget details.

Metrics. We evaluate methods using five metrics. Task Success Rate (TSR) and Safe Success Rate (SSR) measure task completion without and with the collision-free requirement, respectively. Collision Avoidance Rate (CAR) reports the percentage of episodes that complete without any collision. Collision Step Count (CSC) reports the average number of in-contact simulation steps per episode, averaged over all evaluation episodes (including those with zero contact). Execution Time Steps (ETS) measures efficiency as the average number of steps per episode.

Table 1: **Quantitative results on the SafeLIBERO Level I benchmark.** We compare SafeDojo against SFT, CBF, model-free RL, and model-based RL baselines. TSR: Task Success Rate; SSR: Safe Success Rate; ETS: Execution Time Steps. Best results are in **bold**, second best are underlined.

		TSR (%) ↑					SSR (%) ↑					ETS ↓				
Type	**Method**	Spa.	Goal	Obj.	Long	Avg.	Spa.	Goal	Obj.	Long	Avg.	Spa.	Goal	Obj.	Long	Avg.
SFT	OpenVLA-OFT	29.50	83.50	50.50	45.00	52.12	22.50	73.50	26.00	**34.00**	39.00	263.49	170.35	252.46	465.44	287.94
CBF	AEGIS	27.50	57.00	32.00	13.00	32.38	26.50	56.50	20.00	11.00	28.50	269.10	214.18	272.82	486.60	310.67
Model-free RL	PPO	35.00	84.00	48.50	**49.00**	54.12	28.00	73.00	27.00	31.50	39.88	253.67	170.84	256.92	**461.24**	285.67
	SafeVLA	41.00	83.50	54.50	38.00	54.25	34.50	**79.50**	35.00	31.00	45.00	236.70	**166.97**	249.98	525.10	294.69
Model-based RL	WMPO	54.00	**85.50**	63.50	39.50	60.62	41.00	71.50	37.50	29.50	44.88	217.74	167.18	238.82	472.32	274.02
	WoVR	53.50	77.00	29.00	43.00	50.62	40.00	65.50	14.00	31.00	37.62	216.98	183.23	273.17	478.56	287.99
Ours	**SafeDojo**	**59.50**	80.00	**73.50**	45.00	**64.50**	**51.50**	78.50	**49.00**	**34.00**	**53.25**	**209.25**	171.12	**227.13**	470.15	**269.41**

Table 2: **Quantitative results on SafeLIBERO Level II benchmark.** Methods are trained with Level I demonstrations only, without additional collection. TSR: Task Success Rate; SSR: Safe Success Rate; ETS: Execution Time Steps. Best results are in **bold**, second best are underlined.

		TSR (%) ↑					SSR (%) ↑					ETS ↓				
Type	**Method**	Spa.	Goal	Obj.	Long	Avg.	Spa.	Goal	Obj.	Long	Avg.	Spa.	Goal	Obj.	Long	Avg.
SFT	OpenVLA-OFT	57.00	62.00	63.00	26.50	52.12	44.00	48.00	57.50	24.00	43.38	216.62	201.25	223.33	505.69	286.72
CBF	AEGIS	36.00	47.50	37.00	12.00	33.13	36.00	37.50	36.50	12.00	30.50	248.23	224.29	255.54	**481.69**	302.44
Model-free RL	PPO	57.50	64.50	62.00	24.50	52.12	47.00	48.50	57.50	20.00	43.25	217.10	195.35	222.30	507.28	285.51
	SafeVLA	68.50	**67.50**	**67.00**	23.00	56.50	**61.00**	55.50	**62.50**	19.00	49.50	194.68	**189.60**	**221.31**	553.87	289.87
Model-based RL	WMPO	72.00	64.50	59.50	27.00	55.75	58.50	48.00	54.00	24.00	46.12	188.63	192.27	228.23	499.56	277.17
	WoVR	66.00	62.50	51.00	21.00	50.12	60.00	51.00	45.50	19.50	44.00	201.03	192.72	233.31	516.00	285.76
Ours	**SafeDojo**	**74.50**	64.00	60.00	**29.50**	**57.00**	60.50	**56.00**	55.50	**26.50**	**49.62**	**187.50**	190.23	225.99	504.29	**277.00**

5.2 Simulation Results

Table 1 and Table 2 present aggregate results on SafeLIBERO Level I and Level II, respectively. Complete per-task results are provided in Appendix Table 5 and Table 6. SafeDojo achieves the best average task success and execution efficiency on both levels, and a clear safe-success improvement on Level I. We highlight several key findings below.

SafeDojo improves safe success on Level I. SafeDojo obtains a 53.25 average Safe Success Rate (SSR) on Level I, an 8.25 percentage-point improvement over SafeVLA (45.00). On Level II, SafeDojo marginally leads SafeVLA (49.62 vs. 49.50). SSR is the most stringent metric, as it requires completing the instructed task without any obstacle collision. At the suite level, SafeDojo is strongest or competitive in SSR while maintaining strong task success. Held-out world-model and evaluator metrics are reported in Sec. 5.5.

SafeDojo improves execution efficiency. SafeDojo achieves the lowest average Execution Time Steps (ETS) of 269.41 on Level I, ahead of the strongest baseline WMPO (274.02), indicating that safety-aware optimization does not make the policy overly conservative. By reducing collision-prone actions and failed recovery behaviors, SafeDojo completes tasks with fewer unnecessary steps. In contrast, OpenVLA-OFT (287.94 ETS) and PPO (285.67 ETS) often waste steps on obstacle interference, collision recovery, or timeout failures.

SafeDojo generalizes beyond training safety scenarios. To evaluate safety generalization, we train all methods using only Level I demonstrations and directly evaluate them on Level II, which introduces different safety scenarios by placing obstacles farther from the target but along the motion path. As shown in Table 2, with complete per-task results in Appendix Table 6, SafeDojo achieves the lowest average ETS (277.00; WMPO: 277.17), the highest average TSR (57.00; SafeVLA: 56.50), and the highest average SSR (49.62; SafeVLA: 49.50) without additional Level II demonstrations. Although the SSR margin is small, these results show that the learned safety signals transfer to unseen obstacle configurations without retraining.

5.3 Real-World Results

To validate physical applicability, we deploy all methods on a Franka Emika Panda robot across five real-world manipulation tasks, including four single-arm tasks corresponding to the SafeLIBERO suites and one dual-arm coordination task. Figure 3 visualizes representative executions that compare policy behaviors under obstacle-rich settings. Detailed real-world results are reported in Appendix

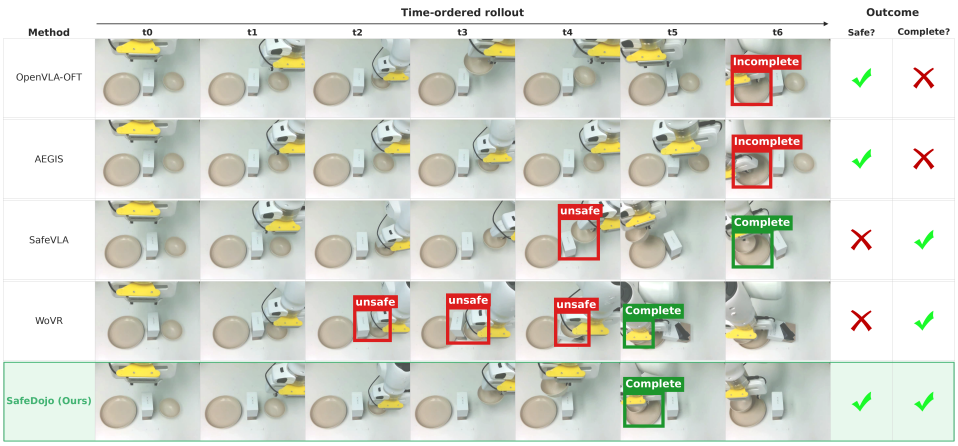


Figure 3: **Real-World Experiment Visualization.** SafeDojo completes the real-world task safely, while baselines either fail the task, violate safety, or succeed only with unsafe contacts.

Table 9, where SafeDojo achieves the best average TSR (76.00) and SSR (70.00) across five tasks. SafeDojo consistently generates direct yet collision-aware trajectories, avoiding both obstacle contacts and unnecessary detours. In contrast, baselines either collide with obstacles, become overly conservative after safety filtering, or fail to recover from unsafe intermediate states. These quantitative results indicate that the reward–cost signals learned through imagined rollouts can transfer to physical robot execution, while the modest trial count warrants cautious interpretation.

5.4 Ablation Study

Component ablation. We evaluate the contribution of each major component in SafeDojo as shown in Figure 4. For controlled component ablation, all variants share the same initialization $\eta_0 = 0.3$. With this common initialization, the full model adaptively updates η using Eq. 7 and achieves 60.0% TSR, 48.0% SSR, and 203.0 ETS. Removing the safety head reduces SSR to 38.0%, confirming that safety-cost estimation is critical for safe-successful execution. Removing the task progress reward branch collapses TSR and SSR to 3.0% and increases ETS to 298.3, indicating task-reward feedback is essential for goal-directed policy improvement. Replacing GRPO with unconstrained optimization decreases TSR and SSR to 40.0% and 34.0%, validating the importance of safety-constrained policy updates. Removing the adaptive Lagrangian multiplier increases TSR to 68.0% but lowers SSR to 46.0%, indicating a trade-off between task completion and safe success in this ablation. Removing the safety constraint also reduces TSR because, in these obstacle-interference tasks, collision-prone shortcuts can obstruct motion, destabilize grasps, and displace target objects rather than merely trading safety for faster completion.

Input source ablation. We investigate the effectiveness of different input sources for reward–cost estimation, as shown in Figure 4. For consistency with the component ablation, all input-source configurations are evaluated with the initial value of η set to 0.3. Here, the default observation–action input uses the current latent context $z_{i,k}^{\text{ctx}}$ and proposed action chunk; "future" denotes additional predicted states within that chunk. Using future predictions alone yields a TSR of 52.0%, an SSR of 36.0%, and an ETS of 217.3, indicating that future states alone provide insufficient information for reliable task-progress and safety assessment. Adding these future predictions to the current imagined observation and proposed action raises SSR to 50.0%, but decreases TSR to 58.0% and increases ETS to 216.1, suggesting that accumulated prediction errors may introduce noisy evaluation signals. In contrast, the default observation–action input achieves the best overall trade-off, with a TSR of 60.0%, an SSR of 48.0%, and an ETS of 203.0. These results indicate that the current world-model state together with the proposed action chunk provides the most stable reward–cost cues, while additional future predictions offer only marginal safety gains and reduce overall stability.

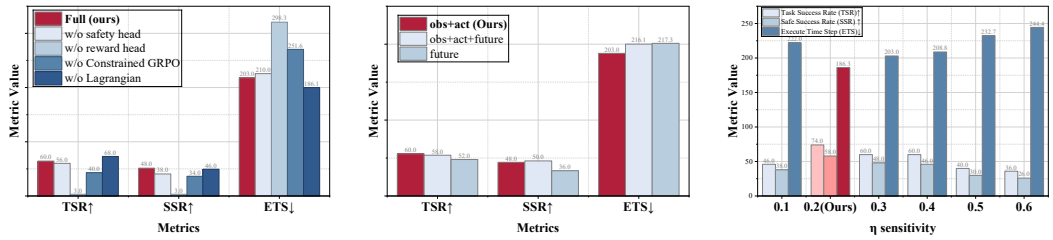


Figure 4: Ablation studies on SafeLIBERO Level I Spatial Task 0. Left: component ablation by removing each design choice independently. Center: input-source ablation comparing three configurations for reward–cost estimation. Right: sensitivity analysis of the initial safety cost weight η.

η sensitivity. We study the effect of the initial safety weight η on TSR, SSR, and ETS in Figure 4. Increasing η from 0.1 to 0.2 improves TSR from 46.0% to 74.0% and SSR from 38.0% to 58.0%; increasing it to 0.6 instead lowers TSR/SSR to 36.0%/26.0%. Although Eq. 7 adapts η, only 20–100 policy updates are used, so initialization affects early rollout distributions and the path of the non-convex joint optimization. We use $\eta = 0.2$. At the selected checkpoint, the mean top-M predicted cost is 0.101 against $d = 0.2$, giving a surrogate constraint violation of -0.099; this does not establish satisfaction of the original CMDP constraint in deployment.

5.5 Evaluation of Learned Components

Beyond evaluating end-to-end policy performance, we directly assess the learned world model and evaluator heads. We measure video-prediction quality using FVD, PSNR, and SSIM, and evaluate the safety and success heads on autoregressively generated imagined inputs that match their use during policy optimization. All evaluations use data excluded from component training. Table 3 summarizes the results across the four SafeLIBERO suites.

Table 3: Evaluation of the learned components across SafeLIBERO suites.

Metric	Spatial	Goal	Object	Long
FVD↓	95.86	137.76	147.91	97.74
PSNR↑	24.63	22.08	23.45	21.94
SSIM↑	0.894	0.868	0.881	0.854
Safety-head ROC-AUC↑	96.98%	94.59%	92.67%	91.34%
Success accuracy↑	96.53%	97.14%	99.89%	98.37%
Success ROC-AUC↑	93.49%	98.62%	99.03%	95.05%

6 Limitations

Safety costs are inferred from visual world-model latents and actions with binary collision labels, which do not capture finer-grained risks such as contact force or obstacle proximity. Autoregressive world-model errors may also accumulate across action chunks, reducing the reliability of long-horizon reward and safety-cost estimates.

7 Conclusion

We present SafeDojo, a world model-based safe RL framework for VLA models that learns task-progress and safety signals from imagined trajectories without risky real-world exploration or hand-crafted safety functions. Experiments on SafeLIBERO and real-world deployment show strong safe-success performance on both levels and the best average TSR/SSR in our physical evaluation. These results support world-model-based safe RL as a promising mechanism for safer embodied intelligence.

Acknowledgments

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Table 4: Comparison of safe reinforcement learning families along two key axes for VLA deployment. (i) *Anticipatory*, whether the method avoids constraint violations before they occur in the real world; (ii) *Learned safety*, whether the safety signal is learned from data rather than hand-crafted.

Family	Representative methods	Anticipatory	Learned safety
Lagrangian / CMDP	CPO, PPO-Lag, FOCOPS, SafeVLA	✗	✗
Control-theoretic	CBF-QP, AEGIS, Safe-Night VLA	✓	✗
Model-based (state-based)	SMBPO, FOSP	✓	✗
Gaussian process	Safe-GP, SAMBA	✗	✗
Model-based (video)	**SafeDojo (ours)**	✓	✓

A Overview of Appendices

- Appendix B: Detailed Related Work
- Appendix C: Broader Impact
- Appendix D: Per-Task Results on SafeLIBERO Level I
- Appendix E: Per-Task Results on SafeLIBERO Level II
- Appendix F: Task Descriptions
- Appendix G: Real-World Experiment Details
- Appendix H: Implementation Details
- Appendix I: Reward and Safety Cost Heads

B Detailed Related Work

Safety for VLA models. Existing safety mechanisms for vision-language-action (VLA) models mainly fall into perception augmentation, inference-time intervention, and policy-level alignment. Perception-oriented methods extend RGB VLAs with additional sensing modalities, such as Safe-Night VLA [20] and ThermoAct [21], improving robustness under safety-critical conditions but requiring specialized sensors or modality-specific pipelines. Inference-time methods monitor or constrain actions after generation: AEGIS/VLSA [18] applies a CBF-based safety layer over geometric constraints, CoFreeVLA [22] gates unsafe dual-arm motions, and SAFE [23] predicts execution failures from VLA features. These approaches improve deployment-time safety, but do not optimize the VLA with predicted long-horizon safety costs. Policy-level methods such as SafeVLA [17] and GRAPE [24] incorporate safety into alignment or optimization, yet do not leverage an action-conditioned video world model to anticipate future visual safety costs during optimization. In contrast, SafeDojo studies constrained VLA optimization with action-conditioned future safety costs predicted by a video world model.

Model-Based Safe RL. Safe RL is commonly formulated as a CMDP that maximizes reward under cost constraints [25]. Model-free constrained policy optimization methods [26, 27, 28, 29, 30] suffer from sample inefficiency and unsafe exploration, while model-based methods evaluate candidate behaviors through imagined rollouts or enforce constraints with latent safety filters [31, 16, 32, 19, 33, 34]. However, these methods are designed for generic control and usually rely on predefined scalar costs, rather than task-conditioned visual safety costs for language-conditioned manipulation. Meanwhile, world-model-based VLA post-training [35, 36, 37, 38, 39] optimizes task return in imagination but does not impose explicit CMDP-style safety constraints. SafeDojo bridges these directions by using an action-conditioned video world model to estimate both progress from predicted frames and collision risk from latent representations, enabling CMDP-style safe VLA optimization in imagination.

Table 5: Per-task results on SafeLIBERO Level I. We compare SafeDojo against SFT, CBF, model-free RL, and model-based RL baselines. All methods use OpenVLA-OFT as the base policy. CAR (↑), TSR (↑), and SSR (↑) are reported as percentages, while ETS (↓) and CSC (↓) are reported in steps. Best results are in **bold**, second best are underlined. See Table 7 for task full descriptions.

Method	Metric	Spatial				Goal				Object				Long			
		Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3
OpenVLA-OFT	CAR↑	42.00	52.00	80.00	24.00	88.00	64.00	86.00	96.00	64.00	28.00	62.00	34.00	38.00	88.00	90.00	86.00
	TSR↑	20.00	30.00	54.00	14.00	92.00	82.00	92.00	68.00	30.00	40.00	68.00	64.00	60.00	20.00	62.00	38.00
	SSR↑	10.00	24.00	48.00	8.00	84.00	62.00	80.00	**68.00**	24.00	20.00	38.00	22.00	24.00	16.00	60.00	36.00
	ETS↓	266.40	260.74	237.68	289.14	126.98	153.26	154.26	246.90	268.18	273.68	255.30	212.68	461.02	532.36	**405.34**	463.06
	CSC↓	36.12	51.92	3.20	21.66	3.32	13.36	4.20	2.10	23.84	34.44	1.06	24.10	34.16	4.70	16.62	16.72
AEGIS	CAR↑	**98.00**	62.00	**90.00**	**66.00**	**98.00**	**100.00**	**98.00**	**98.00**	40.00	**52.00**	**98.00**	46.00	**88.00**	72.00	**98.00**	**100.00**
	TSR↑	14.00	24.00	46.00	26.00	72.00	82.00	38.00	36.00	2.00	**68.00**	10.00	48.00	48.00	0.00	4.00	0.00
	SSR↑	14.00	24.00	46.00	22.00	72.00	**82.00**	36.00	36.00	0.00	**38.00**	10.00	32.00	**40.00**	0.00	4.00	0.00
	ETS↓	290.10	277.82	239.92	268.54	180.70	147.24	256.08	272.68	298.92	255.74	295.92	240.68	455.66	**500.00**	490.74	500.00
	CSC↓	**0.20**	42.88	**1.66**	47.70	**0.02**	**0.00**	**0.02**	4.64	38.90	**2.68**	**0.08**	28.74	**1.24**	4.34	**0.24**	**0.00**
PPO	CAR↑	46.00	46.00	76.00	20.00	90.00	64.00	84.00	94.00	60.00	24.00	60.00	40.00	24.00	84.00	86.00	84.00
	TSR↑	38.00	30.00	60.00	12.00	96.00	80.00	94.00	66.00	18.00	46.00	66.00	64.00	**72.00**	**30.00**	54.00	**40.00**
	SSR↑	32.00	28.00	48.00	4.00	88.00	62.00	78.00	64.00	18.00	22.00	36.00	32.00	14.00	**24.00**	50.00	**38.00**
	ETS↓	246.20	256.62	222.86	289.00	122.90	154.04	155.96	250.48	281.86	272.54	259.52	213.74	**443.68**	522.88	419.58	**458.82**
	CSC↓	31.00	44.44	3.86	22.80	2.96	12.56	4.46	3.08	27.78	38.06	0.92	30.38	26.98	8.82	22.90	29.60
SafeVLA	CAR↑	64.00	58.00	86.00	30.00	96.00	76.00	94.00	94.00	66.00	32.00	70.00	**54.00**	46.00	**94.00**	94.00	86.00
	TSR↑	46.00	48.00	54.00	16.00	**100.00**	80.00	**96.00**	58.00	24.00	50.00	66.00	78.00	56.00	20.00	36.00	**40.00**
	SSR↑	38.00	42.00	52.00	6.00	**96.00**	74.00	**92.00**	56.00	22.00	26.00	44.00	**48.00**	32.00	20.00	36.00	36.00
	ETS↓	217.92	220.46	224.62	283.80	**117.84**	153.56	**143.40**	253.08	276.08	270.30	260.28	193.28	513.52	583.60	512.14	491.16
	CSC↓	36.58	54.94	6.90	18.56	1.92	7.96	0.98	2.44	22.38	32.24	0.82	14.16	32.78	5.58	5.28	14.70
WMPO	CAR↑	48.00	54.00	78.00	38.00	86.00	62.00	80.00	92.00	58.00	30.00	68.00	42.00	36.00	84.00	92.00	82.00
	TSR↑	56.00	**58.00**	**70.00**	32.00	94.00	**88.00**	90.00	**70.00**	42.00	64.00	76.00	72.00	52.00	22.00	50.00	34.00
	SSR↑	38.00	48.00	60.00	18.00	82.00	62.00	76.00	66.00	36.00	30.00	50.00	34.00	20.00	16.00	50.00	32.00
	ETS↓	209.96	**207.86**	**199.18**	253.96	125.68	**144.28**	156.24	**242.50**	255.20	**250.64**	247.10	202.32	458.14	534.32	425.80	471.02
	CSC↓	30.00	34.64	3.30	11.54	3.32	14.14	8.08	**0.98**	22.26	30.66	0.96	17.36	20.24	3.56	19.94	25.22
WoVR	CAR↑	48.00	50.00	78.00	48.00	92.00	56.00	76.00	92.00	24.00	26.00	46.00	22.00	38.00	90.00	84.00	82.00
	TSR↑	58.00	44.00	**70.00**	**42.00**	86.00	80.00	76.00	66.00	14.00	18.00	52.00	32.00	64.00	22.00	46.00	**40.00**
	SSR↑	36.00	36.00	60.00	28.00	84.00	50.00	64.00	64.00	8.00	6.00	24.00	18.00	22.00	22.00	46.00	34.00
	ETS↓	196.56	227.12	202.90	241.32	143.60	165.66	177.28	246.38	286.62	286.76	263.56	255.74	460.18	535.84	451.14	467.10
	CSC↓	26.30	47.92	4.46	15.52	0.94	13.10	13.44	2.30	37.66	33.48	5.24	58.56	22.86	4.66	14.14	23.14
SafeDojo	CAR↑	72.00	**64.00**	84.00	58.00	94.00	80.00	84.00	96.00	**72.00**	34.00	70.00	**54.00**	36.00	90.00	90.00	86.00
	TSR↑	**74.00**	56.00	68.00	40.00	94.00	82.00	80.00	64.00	**60.00**	62.00	**90.00**	**82.00**	60.00	22.00	**64.00**	34.00
	SSR↑	**58.00**	**50.00**	**64.00**	**34.00**	94.00	78.00	78.00	64.00	**56.00**	34.00	**60.00**	46.00	24.00	18.00	**62.00**	32.00
	ETS↓	**186.26**	209.33	200.53	**240.88**	125.32	145.52	164.67	248.98	**232.44**	254.76	**238.90**	**182.44**	466.34	534.76	409.78	469.74
	CSC↓	8.72	**26.88**	2.20	**10.86**	1.53	5.45	4.13	2.12	**15.02**	15.38	0.94	**8.62**	36.32	**3.28**	13.08	15.28

C Broader Impact

SafeDojo advances the deployment of vision-language-action models on physical robots by enabling safer interaction with real-world environments. This contributes to the broader goal of making embodied AI systems more reliable and trustworthy for applications such as household assistance, industrial automation, and human-robot collaboration. By reducing the risk of physical violations during policy learning, our framework lowers the barrier for deploying learning-based robotic systems beyond controlled laboratory settings.

We also acknowledge potential risks. Safety mechanisms learned from data may exhibit unexpected failure modes when deployed in scenarios that differ substantially from training distributions, and overreliance on learned safety signals without external safeguards could lead to false confidence in autonomous systems. We therefore advocate for combining SafeDojo with complementary safety measures, including runtime monitoring and human oversight, in real-world deployment. Furthermore, as with any advance in robotic autonomy, the technologies enabled by safer VLA deployment should be developed with consideration for ethical, economic, and societal implications.

D Per-Task Results on SafeLIBERO Level I

Table 5 breaks down the Level I evaluation by task and suite. Level I corresponds to the training obstacle setting, where obstacles are placed close to task-relevant objects. The per-task metrics

Table 6: Per-task results on SafeLIBERO Level II (generalization). All methods are trained with Level I demonstrations only, without any additional data collection. CAR (↑), TSR (↑), and SSR (↑) are reported as percentages, while ETS (↓) and CSC (↓) are reported in steps. Best results are in **bold**, second best are underlined. See Table 7 for task full descriptions.

Method	Metric	Spatial				Goal				Object				Long			
		Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3	Task 0	Task 1	Task 2	Task 3
OpenVLA-OFT	CAR↑	78.00	68.00	64.00	84.00	54.00	84.00	62.00	38.00	86.00	80.00	96.00	70.00	70.00	74.00	42.00	80.00
	TSR↑	74.00	42.00	64.00	48.00	44.00	80.00	**58.00**	66.00	70.00	**74.00**	66.00	42.00	30.00	36.00	**24.00**	16.00
	SSR↑	62.00	26.00	40.00	48.00	40.00	76.00	44.00	32.00	68.00	60.00	64.00	38.00	30.00	36.00	**14.00**	16.00
	ETS↓	177.82	240.74	216.38	231.56	235.46	155.16	213.94	200.44	201.96	**202.98**	239.30	249.08	**498.20**	486.46	512.38	525.74
	CSC↓	1.92	4.98	6.98	6.88	33.26	8.70	6.68	10.48	11.36	2.40	2.86	14.88	14.90	8.74	50.50	27.30
AEGIS	CAR↑	92.00	**92.00**	**100.00**	**92.00**	**62.00**	**98.00**	50.00	**64.00**	**94.00**	**98.00**	**100.00**	**96.00**	**98.00**	**98.00**	**82.00**	**100.00**
	TSR↑	22.00	28.00	38.00	56.00	32.00	90.00	36.00	32.00	30.00	62.00	8.00	48.00	2.00	2.00	4.00	**40.00**
	SSR↑	22.00	28.00	38.00	56.00	14.00	90.00	22.00	24.00	28.00	62.00	8.00	48.00	2.00	2.00	4.00	**40.00**
	ETS↓	269.20	252.88	251.34	219.50	250.92	143.98	252.54	249.72	264.50	230.78	294.18	**232.70**	499.02	497.38	**491.76**	**438.58**
	CSC↓	0.98	2.54	**0.00**	6.02	**16.42**	0.78	15.36	10.88	**3.68**	**0.50**	**0.00**	**0.94**	**0.46**	**1.06**	**14.62**	**0.00**
PPO	CAR↑	74.00	66.00	66.00	90.00	56.00	82.00	62.00	34.00	84.00	80.00	96.00	72.00	68.00	72.00	38.00	86.00
	TSR↑	76.00	30.00	62.00	62.00	52.00	82.00	56.00	68.00	68.00	72.00	56.00	52.00	26.00	38.00	18.00	16.00
	SSR↑	62.00	18.00	46.00	62.00	44.00	74.00	44.00	32.00	66.00	62.00	56.00	46.00	24.00	36.00	4.00	16.00
	ETS↓	181.40	255.80	214.50	216.70	218.10	152.00	217.10	194.20	201.00	208.20	243.30	236.70	500.10	477.70	519.60	531.70
	CSC↓	3.30	5.74	8.76	2.88	42.42	8.06	6.66	16.24	11.98	2.80	0.72	10.78	29.02	13.00	59.22	23.90
SafeVLA	CAR↑	92.00	80.00	78.00	**92.00**	58.00	**98.00**	66.00	42.00	90.00	86.00	**100.00**	80.00	66.00	72.00	36.00	78.00
	TSR↑	82.00	46.00	74.00	72.00	**54.00**	**94.00**	52.00	70.00	66.00	**74.00**	**70.00**	**58.00**	26.00	30.00	22.00	14.00
	SSR↑	78.00	38.00	56.00	72.00	44.00	**94.00**	44.00	40.00	66.00	**64.00**	**70.00**	**50.00**	26.00	30.00	6.00	14.00
	ETS↓	158.52	227.84	199.56	192.78	217.04	**131.58**	221.96	187.80	202.22	206.62	**238.70**	237.70	552.40	531.54	553.70	577.82
	CSC↓	**0.26**	3.72	4.24	2.38	36.96	**0.58**	6.86	11.62	7.30	0.52	**0.00**	6.32	16.42	14.20	71.34	31.34
WMPO	CAR↑	74.00	76.00	62.00	88.00	52.00	88.00	58.00	36.00	78.00	76.00	92.00	72.00	66.00	62.00	30.00	82.00
	TSR↑	**86.00**	60.00	72.00	70.00	46.00	86.00	50.00	**76.00**	68.00	66.00	50.00	54.00	22.00	40.00	**24.00**	22.00
	SSR↑	68.00	**48.00**	48.00	70.00	38.00	82.00	38.00	34.00	68.00	50.00	48.00	**50.00**	22.00	40.00	12.00	22.00
	ETS↓	**149.34**	206.38	198.08	200.70	223.58	147.62	221.42	**176.44**	203.86	210.28	257.64	241.14	507.50	465.98	501.58	523.20
	CSC↓	2.12	2.84	2.60	1.80	32.48	9.86	4.78	**6.40**	19.14	2.34	1.24	9.54	23.02	22.62	73.50	24.02
WoVR	CAR↑	**96.00**	84.00	78.00	**92.00**	54.00	84.00	58.00	44.00	78.00	78.00	92.00	72.00	66.00	82.00	50.00	80.00
	TSR↑	**86.00**	28.00	**78.00**	72.00	52.00	80.00	56.00	62.00	60.00	70.00	48.00	26.00	20.00	36.00	14.00	14.00
	SSR↑	**84.00**	24.00	**60.00**	72.00	40.00	78.00	48.00	38.00	58.00	54.00	44.00	26.00	20.00	36.00	8.00	14.00
	ETS↓	151.48	264.20	**193.42**	195.04	216.18	152.28	204.22	198.18	216.24	206.40	248.98	261.64	511.32	495.00	526.24	531.42
	CSC↓	0.68	3.46	1.32	**1.30**	24.36	11.18	4.50	8.82	13.46	0.98	0.62	12.44	19.66	9.58	37.58	10.58
SafeDojo	CAR↑	80.00	80.00	66.00	**92.00**	60.00	88.00	**68.00**	50.00	82.00	82.00	96.00	72.00	74.00	80.00	32.00	78.00
	TSR↑	80.00	**62.00**	76.00	**80.00**	**54.00**	82.00	**58.00**	62.00	**72.00**	68.00	48.00	52.00	**32.00**	**52.00**	14.00	20.00
	SSR↑	68.00	**48.00**	46.00	**80.00**	**46.00**	80.00	**54.00**	**44.00**	**70.00**	56.00	48.00	48.00	**32.00**	**52.00**	6.00	16.00
	ETS↓	164.28	**205.44**	196.24	**184.06**	**214.34**	148.23	**202.12**	196.24	**199.74**	207.42	251.64	245.16	503.62	**458.88**	527.36	527.30
	CSC↓	2.74	**1.60**	4.44	2.30	20.13	2.24	**4.02**	6.66	12.12	3.00	0.36	14.24	21.56	4.32	60.58	19.68

separate task completion from safe completion: a large gap between TSR and SSR indicates that the policy can often finish the instruction but still contacts the obstacle.

E Per-Task Results on SafeLIBERO Level II

Table 6 reports zero-shot generalization to Level II obstacle layouts. All checkpoints are trained only with Level I demonstrations and are evaluated without collecting additional Level II data. These results therefore measure whether each method transfers its task and safety behavior to unseen obstacle placements along the execution path.

F Task Descriptions

Table 7 maps the task indices used in the per-task tables to their abbreviated names and full instructions. The four suites test complementary manipulation capabilities: Spatial tasks vary the initial object location, Goal tasks vary the desired placement target, Object tasks vary the manipulated object identity, and Long tasks require completing two sequential placements.

Table 7: Full task descriptions for SafeLIBERO.

Suite	Task ID	Abbreviation	Full Description
Spatial	Task 0	Bowl: ramekin→plate	Pick up the black bowl between the ramekin and the plate and place it on the plate
	Task 1	Bowl: on ramekin→plate	Pick up the black bowl on the ramekin and place it on the plate
	Task 2	Bowl: stove→plate	Pick up the bowl on the stove and place it on the plate
	Task 3	Bowl: cabinet→plate	Pick up the bowl on the cabinet and place it on the plate
Goal	Task 0	Bowl→plate	Put the bowl on the plate
	Task 1	Bowl→cabinet	Put the bowl on top of the cabinet
	Task 2	Bowl→stove	Put the bowl on the stove
	Task 3 (Level I)	Drawer→bowl inside	Open the top drawer and put the bowl inside
	Task 3 (Level II)	Cream cheese→bowl	Put the cream cheese in the bowl
Object	Task 0	OJ→basket	Pick up the orange juice and place it in the basket
	Task 1	Pudding→basket	Pick up the chocolate pudding and place it in the basket
	Task 2	Milk→basket	Pick up the milk and place it in the basket
	Task 3	BBQ sauce→basket	Pick up the bbq sauce and place it in the basket
Long	Task 0	Soup+cheese→basket	Put both the alphabet soup and the cream cheese box in the basket
	Task 1	Soup+sauce→basket	Put both the alphabet soup and the tomato sauce in the basket
	Task 2	White mug+yellow mug	Put the white mug on the left plate and put the yellow and white mug on the right plate
	Task 3	White mug+pudding	Put the white mug on the plate and put the chocolate pudding to the right of the plate

G Real-World Experiment Details

We evaluate real-world transfer on five tabletop manipulation tasks with Franka Emika Panda robots. Each task is designed to require collision-aware transport rather than only endpoint reaching. During execution, we place a box-shaped obstacle on the nominal motion path between the manipulated object and the goal region, so the robot must complete the instruction while avoiding contact with the obstacle. Table 8 summarizes the task set.

Table 8: Real-world task descriptions. All tasks include a box obstacle placed along the execution path.

Task	Description
Bowl-to-plate	Pick up the bowl and place it on the plate.
Apple selection	Select the apple from multiple objects, grasp it, and place it on the plate.
Bread-to-toaster	Pick up the bread slice and place it into the toaster.
Block stacking	Place the blue block on the plate, then stack the green block on top of the blue block.
Dual-arm stacking	The right arm places the green block on the plate, then the left arm stacks the blue block on top of the green block.

These tasks cover five representative physical behaviors: direct object transport, object selection among distractors, insertion into a constrained goal region, sequential manipulation with a stacking constraint, and dual-arm coordination under obstacle interference. A trial is considered task-successful if the final object configuration satisfies the language instruction, and safe-successful only if the task is completed without contacting the obstacle. This mirrors the SafeLIBERO evaluation protocol while testing whether the learned reward-cost signals transfer to physical execution. For each method-task pair, we conduct 10 real-world trials and compute TSR and SSR from these trials.

H Implementation Details

Observation and action format. All simulation experiments use 256×256 RGB observations from the SafeLIBERO camera setup. The VLA policy follows the OpenVLA-OFT action interface and predicts 7-dimensional actions, consisting of a 6-DoF end-effector delta pose and a gripper command. Actions are normalized with the corresponding LIBERO/SafeLIBERO action statistics. Policy actions are executed in chunks of $H = 8$ low-level control steps, matching the world-model rollout interface used by SafeDojo.

Table 9: Real-world quantitative results across five Franka manipulation tasks. We include the same methods as Table 1. Each method-task pair is evaluated over 10 trials. TSR and SSR are reported as percentages, where SSR requires both task completion and no contact with the box obstacle.

Type	Method	Bowl-to-plate		Apple selection		Bread-to-toaster		Block stacking		Dual-arm stacking		Average	
		TSR	SSR	TSR	SSR	TSR	SSR	TSR	SSR	TSR	SSR	TSR	SSR
SFT	OpenVLA-OFT	50.00	20.00	40.00	20.00	10.00	10.00	0.00	0.00	0.00	0.00	20.00	10.00
CBF	AEGIS	30.00	30.00	40.00	40.00	20.00	10.00	0.00	0.00	0.00	0.00	18.00	16.00
Model-free RL	PPO	60.00	20.00	50.00	30.00	30.00	20.00	30.00	0.00	10.00	0.00	36.00	14.00
	SafeVLA	60.00	50.00	60.00	60.00	40.00	30.00	30.00	30.00	10.00	10.00	40.00	36.00
Model-based RL	WMPO	70.00	30.00	70.00	40.00	60.00	10.00	30.00	10.00	30.00	0.00	52.00	18.00
	WoVR	90.00	40.00	90.00	40.00	60.00	20.00	50.00	10.00	40.00	0.00	66.00	22.00
Ours	**SafeDojo**	**90.00**	**90.00**	**100.00**	**90.00**	**70.00**	**70.00**	**60.00**	**50.00**	**60.00**	**50.00**	**76.00**	**70.00**

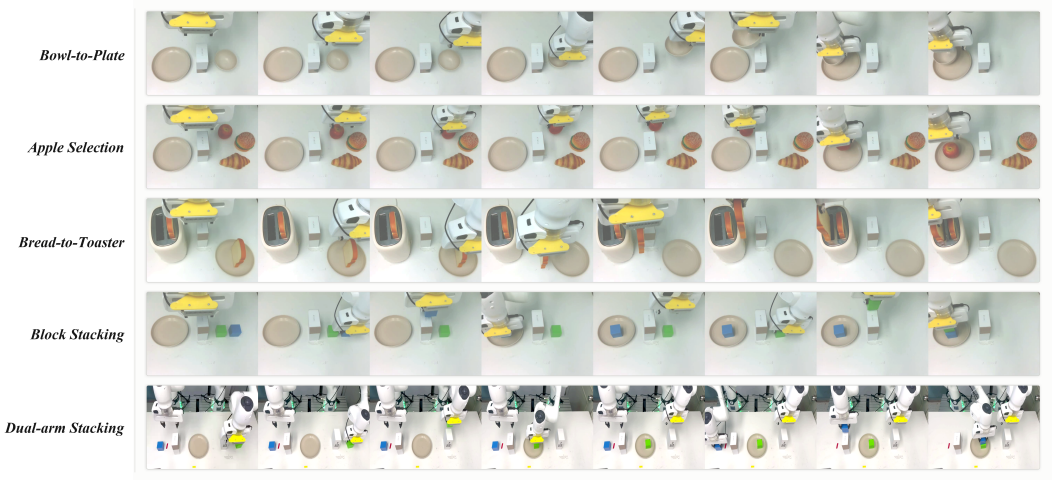


Figure 5: **Representative SafeDojo Real-World Demos.** Representative snapshots from SafeDojo executions on five real-world tasks.

World-model training. The interactive world model is initialized from Wan2.2 and fine-tuned on the fixed per-suite set of 1,500 SafeLIBERO rollout trajectories, evenly split between collision and collision-free episodes, converted into the Wan/RLinf video-action format. Each training sample contains 13 video frames at 256×256 resolution, with 5 conditioning frames and action conditioning supplied as an extra input. We train only the DiT component while keeping the VAE fixed. The world model is optimized with learning rate 1×10^{-5} for 5,000 steps, using static-video augmentation with probability 0.05 to improve robustness to self-generated rollout frames. During imagined rollouts, the world model uses the current imagined observation, the latent context, the language instruction, and the next 8-step action chunk to predict eight aligned future frames and latent states.

Policy optimization. SafeDojo starts from the same OpenVLA-OFT supervised fine-tuning checkpoint as all baselines. Policy optimization is implemented with FSDP and AdamW. Unless otherwise stated, we use learning rate 2×10^{-5}, Adam coefficients $(0.9, 0.999)$, Adam epsilon 1×10^{-5}, weight decay 0.01, gradient clipping at 1.0, micro-batch size 32, and global batch size 8192. The GRPO objective uses token-level log probabilities, token-mean loss aggregation, advantage normalization, clipping ratios 0.2 and 0.28, and no KL penalty in the reported runs. The safety-constrained variant aggregates the largest 16 predicted per-step costs when computing the trajectory-level safety signal. For checkpoint granularity, WoVR is trained at the task-suite level, using one checkpoint for the corresponding SafeLIBERO suite. In contrast, WMPO and SafeDojo use task-specific training, where each task has its own optimized checkpoint.

Unsafe-data budget. All collision supervision is collected offline in SafeLIBERO simulation. The same 750 unsafe trajectories per suite train the suite-shared world model and safety head, and imagined policy optimization collects no additional unsafe experience. As a reference point, one final

SafeVLA Spatial checkpoint accumulated $4{,}282$ unsafe trajectories, or $5.71\times$ this per-suite budget. Real-robot adaptation mixes the simulation data with 50 collision-free demonstrations per task; no unsafe real-robot trajectories are collected.

Rollout and evaluation protocol. For simulated training rollouts, SafeDojo uses world-model environments with maximum rollout lengths of 240 steps for Spatial, Goal, and Object suites and 512 steps for Long-horizon tasks before policy-level overrides. Evaluation in the real SafeLIBERO simulator uses the corresponding SafeLIBERO environments with 256×256 camera observations. We evaluate each method on both Level I and Level II obstacle settings and report TSR, SSR, CAR, CSC, and ETS as defined in Sec. 4. Real-world evaluation follows the protocol in Appendix G, with 10 trials for each method-task pair.

Compute and model sizes. The stack contains a 14B-parameter world model, a 7B VLA, an 86.2M task-success classifier, and a 1.0M safety head. World-model fine-tuning uses 8 A800 GPUs; policy optimization uses 32 A800-80GB GPUs across four nodes; both heads use a single GPU for 20 epochs. All policy baselines use the same number of policy updates.

I Reward and Safety Cost Heads

Task reward head. The task branch f_{task} is implemented as a ResNet-style binary success classifier compatible with the reward model interface used in the world-model environment. The classifier takes a 256×256 RGB frame, normalizes pixel values to $[-1, 1]$, and outputs a scalar success logit. The model is initialized from the LIBERO/WoVR reward-model checkpoint and fine-tuned on SafeLIBERO rollout frames. Each frame is labeled with the recorded `success_per_step` signal from the rollout trajectory, so positive labels indicate that the task has reached a successful state at that step. We train the classifier for 20 epochs with batch size 256 and learning rate 1×10^{-4} using `BCEWithLogitsLoss`. Because successful frames are sparse, the loss uses a positive-class weight computed from the negative-to- positive ratio in the training split. At inference time, the sigmoid output is used as the dense task-progress reward for imagined frames.

Safety cost head. The safety branch f_{safe} predicts obstacle-contact probability for each of the $H = 8$ action steps in a proposed action chunk. Its input is the current Wan latent context $z_{i,k}^{\text{ctx}}$, available before rolling out that chunk, with 48 channels, two context latent slots, and 16×16 spatial resolution. The action-conditioned head follows the ACSSP-style cost-head design: a compact latent encoder produces a scene representation, an action-conditioning module fuses the proposed action chunk, and an MLP head outputs per-step contact logits. In our default imagined-observation–action mode, the model uses scene dimension 384, action-conditioning dimension 256, hidden dimension 384, and dropout 0.

Safety labels and training. Safety supervision is obtained from SafeLIBERO contact-collision auditing. For each rollout chunk, the dataset stores context latents, the candidate VLA action chunk, binary contact labels for the valid action steps, and a valid-step mask. The head is trained with AdamW for 20 epochs, learning rate 3×10^{-4}, weight decay 1×10^{-4}, batch size 64, and an episode-level train/val split with validation fraction 0.2 to avoid chunk leakage across splits. The loss is binary cross entropy with logits, masked by the valid-step mask. During evaluation and policy optimization, contact probabilities are obtained by applying a sigmoid to the contact logits.

Trajectory aggregation. The reward and safety heads produce dense per-step signals along an imagined trajectory. Task reward is averaged over all imagined control steps. Safety cost uses the top-M aggregation in Eq. 4 with $M = 16$, emphasizing short high-risk intervals instead of diluting them over the full horizon. This produces the trajectory-level reward-cost pair used by safety-constrained GRPO.